

Practical Machine Learning and Deep Learning

D1.2 Progress Submission

Team "Magic Photo"

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October 7, 2025

1 Project Topic

This project objective is to enhance existing solutions for image restoration from different corruptions and coloring of grayscale images using machine learning and deep learning techniques. The project will focus on developing a user-friendly application that allows users to upload images, select the processing option they want to apply, and receive the processed image as output. Additionally, we plan to collect users' feedback for future product improvements and keep track of software lifecycle. The application will leverage pre-trained models and fine-tune them.

2 Link to the Repository

The link: <https://github.com/ivaniinno/Image-Restoration-and-Coloring/tree/dev>. For now we store code in **dev** branch with **data_augmentation** folder, **docs** folder with project documentation, **models_testing** folder for trying baseline model implementations, and in **models_finetuning** we have a template for finetuning. Branch **issue-2** contains exploratory data analysis.

3 Dataset and Exploratory Data Analysis

Our datasets are composed from three sources:

- Nature Dataset: <https://huggingface.co/datasets/mertcobanov/nature-dataset> (**nature** category in our dataset)
- European Cities 1M dataset: <http://image.ntua.gr/iva/datasets/ec1m/index.html> (**city** category in our dataset)
- CelebA Dataset: <https://www.kaggle.com/datasets/jessicali9530/celeba-dataset> (**faces** category in our dataset)

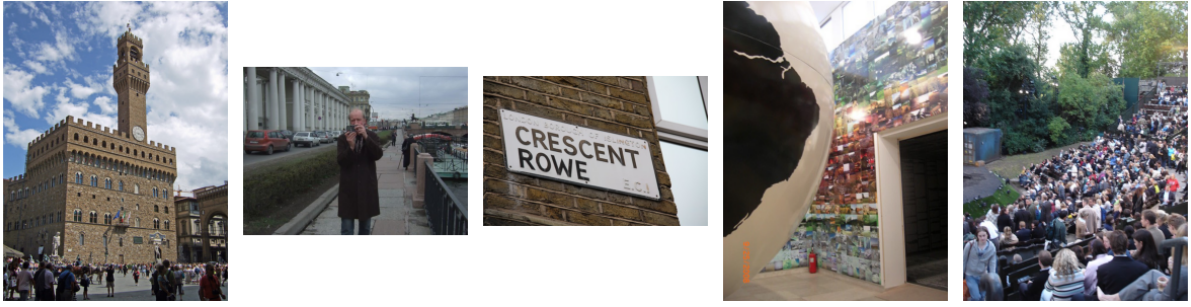
One can find image examples in Figure 1

To balance category distribution, we randomly sampled 50000 images from each source.

Results of data exploration:

1. 56 duplicates were found and removed
2. 2743 gray images were deleted because they are not suitable for the colorization task (we cannot compare the result with the original image) as shown on Figure 2.
3. **faces** category has resolution of 178x218, **nature** category has resolution of 512x512, and **city** category resolutions varies as Figures 3 and 4 demonstrate. Figure 4 shows that most of the images have aspect ratio 1-2. Images with ratios more than 6 were removed as extreme cases (in total 24 of them).

Category: city



Category: faces



Category: nature



Figure 1: Images Examples from Each Category

4. Images were divided into classes via ResNet (for feature extraction), and PCA was used for visualization (see Figure 5). In general, classes are well separable, however, images from the **city** category slightly overlap with two other clusters, which indicates the presence of common visual patterns between the **city** category and the other two classes.
5. Average brightness distribution is as shown on Figure 6. 1457 outliers ($\approx 1\%$) were found (underexposed and overexposed images) (see Figure 7). These images were saved and will be used for training the model on hard cases.
6. As a result, 147177 images remain from 150000:
 - city - 48112
 - faces - 49985
 - nature - 49080

The final classes proportion seems to be appropriate.

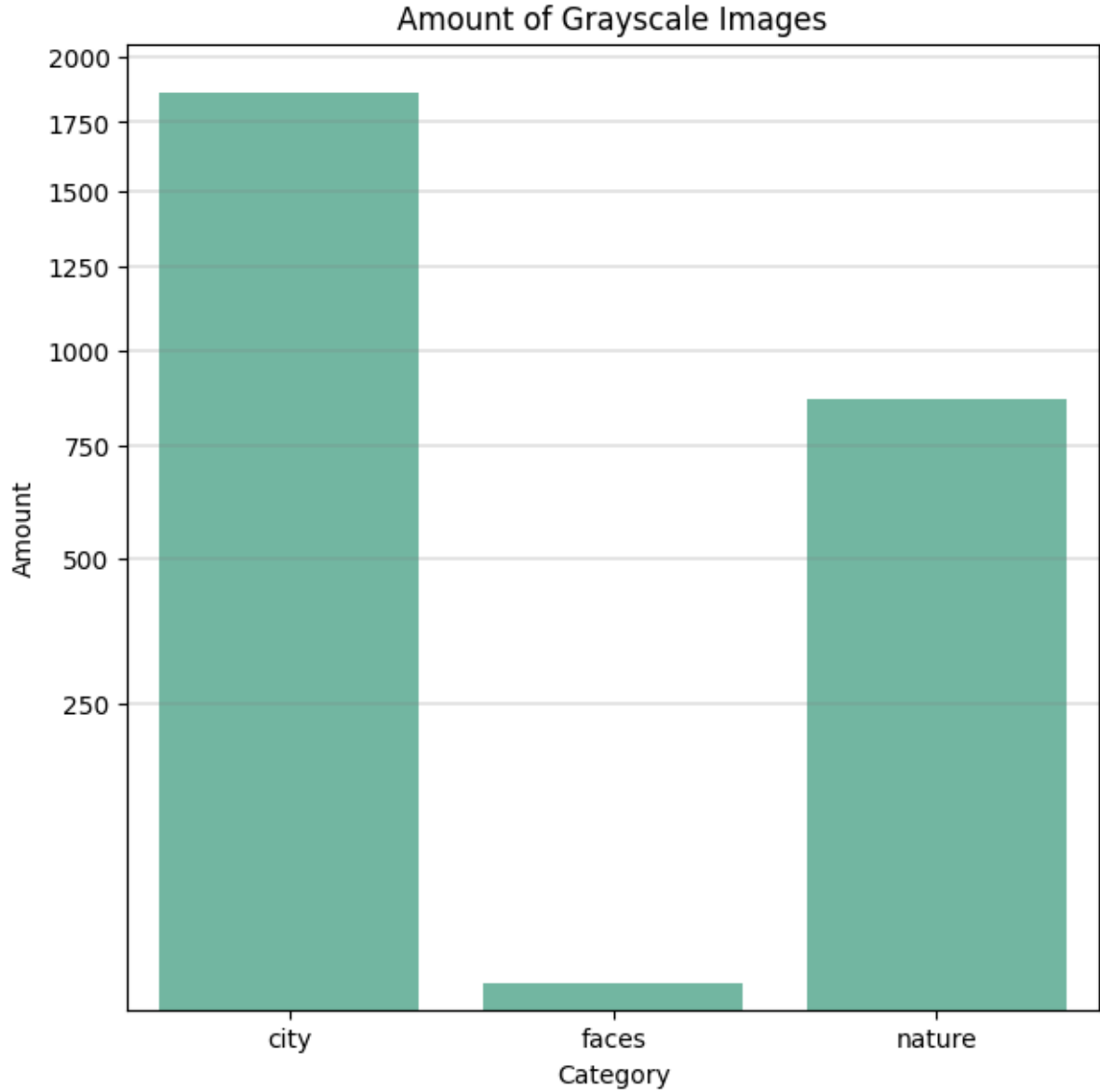


Figure 2: Amount of Gray Images in Each Category

4 Data Augmentation

Several data augmentation techniques were implemented to increase dataset size and variability.

Degradation Types include:

1. Noise: Gaussian, salt-pepper, speckle
2. Blur: Gaussian, median, motion blur
3. Resolution: Downscaling with upscaling artifacts
4. Compression: JPEG artifacts (quality 20-30)
5. Physical damage: Scratches and dust spots
6. Color: Grayscale conversion

Additionally, mentioned degradations were combined to create more complex scenarios as follows:

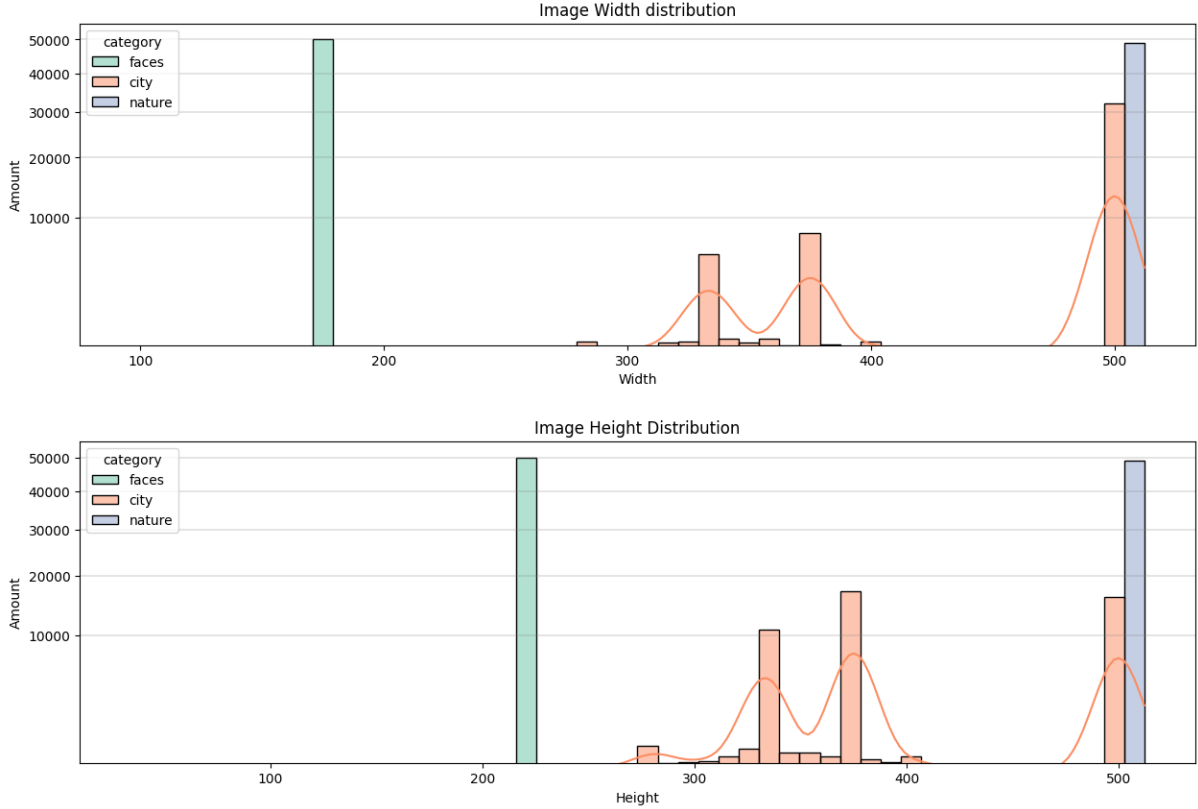


Figure 3: Images Height and Width Distribution

1. **combined_restoration:** Dust_and_Scratch/None + noise(Gaussian/SP/Speckle) + low resolution/compression
2. **combined_superres:** Blur(Gaussian/Median/Motion blur) + resolution reduction/compression
3. **combined_colorization:** Grayscale + Noise(Gaussian/SP/Speckle)/Blur(Gaussian/Median/Motion blur)/None

5 Model Testing

Real-ESRGAN and GPFGAN models were tested out-of-the-box on selected images for future comparisons after fine-tuning. The notebook is located at **models_testing/restoration_superres/restorationtest**. Preferable link to open the notebook: <https://www.kaggle.com/code/railsabirov/restorationtest>.

The results are on Figure 8.

As one can see, on average PSNR shows 24-27 dB, and SSIM shows 0.74-0.81. Such metrics show that the restored image is quite good in quality. The combination of Real-ESRGAN + GPFGAN works best for restoring images with faces, then comes the urban landscape, and it works worst with the nature category, since there is a lot of noise initially in the image, which the model smooths out.

6 Future Work

In the closest perspective we are planning from highest to lowest priority (first 2 points are crucial and to be done first):

- Fine-tune Real-ESRGAN and GPFGAN models.
- Investigate other models for possible consideration of them.
- Start developing simple backend/frontend for interaction with model.

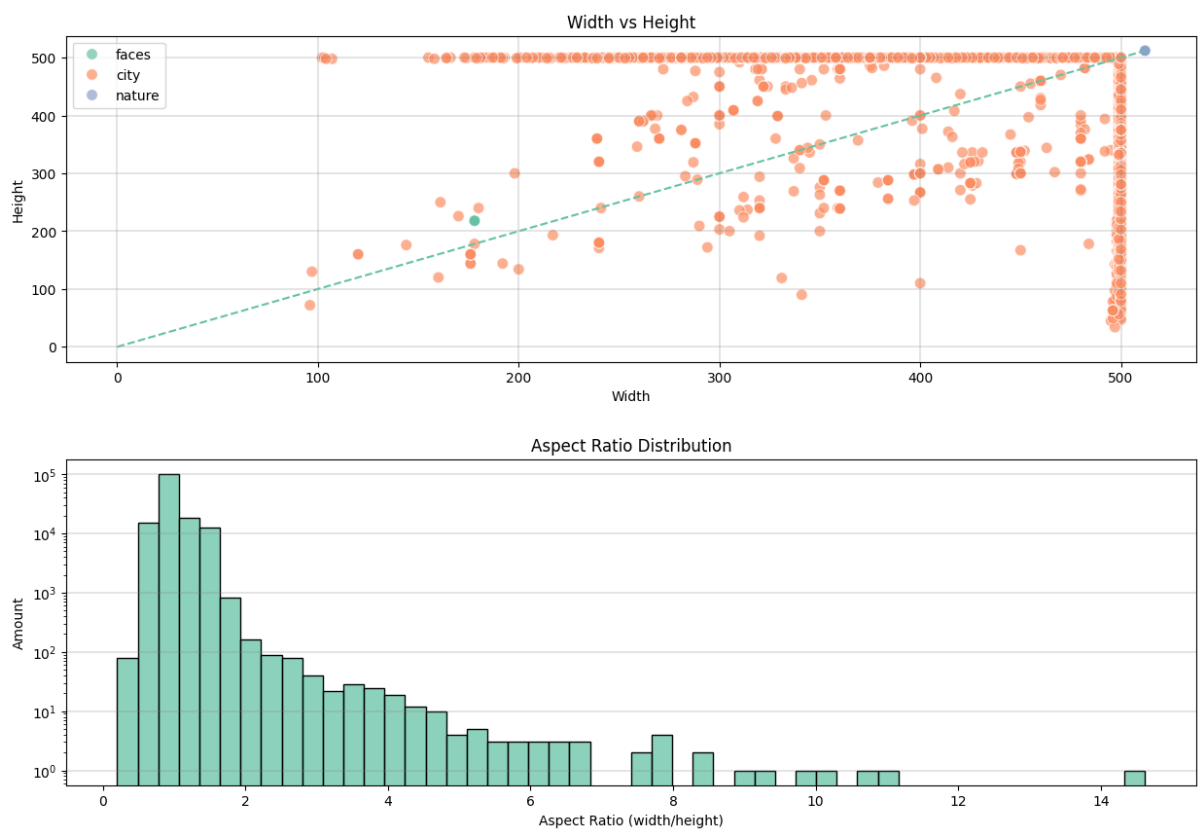


Figure 4: Images Height and Width Comparison

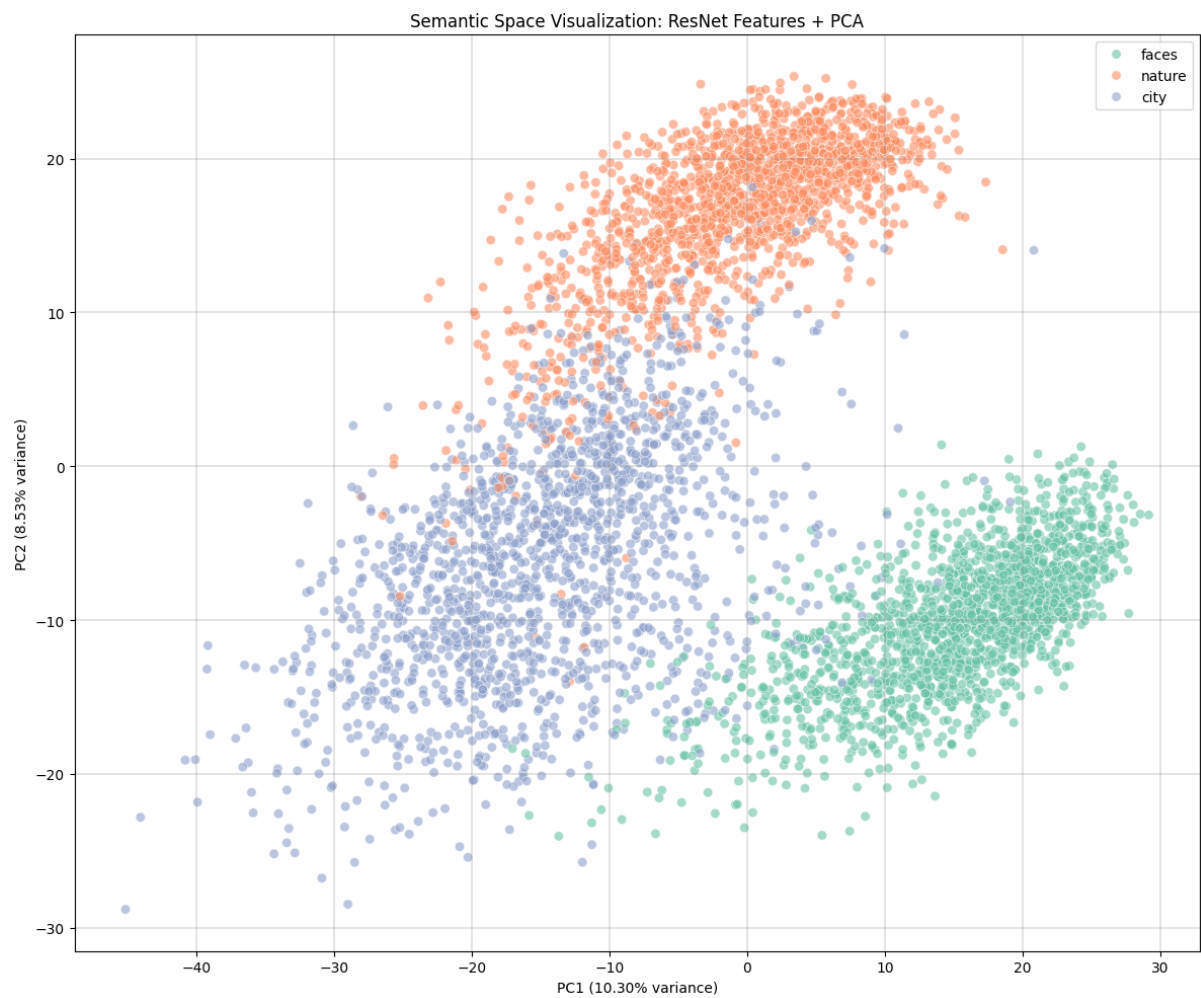


Figure 5: Image Features and Classes Visualization

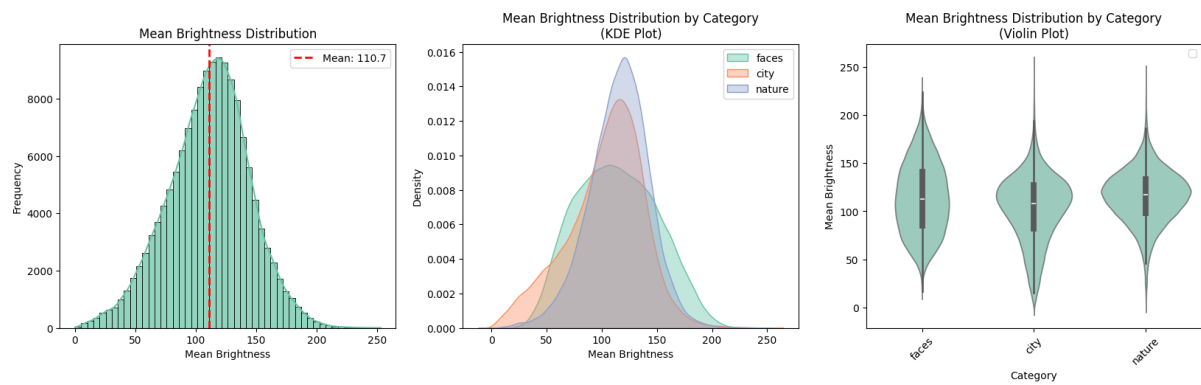


Figure 6: Images Brightness Distribution

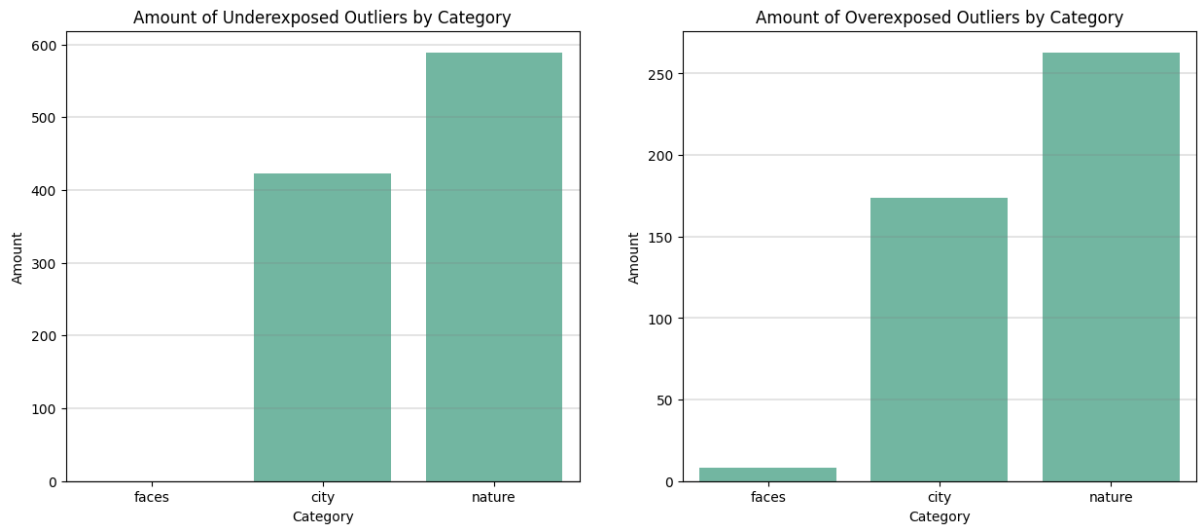


Figure 7: Underexposed and Overexposed Images Counts

```
print(summary)
```

[14] Python

...	category	type	PSNR	SSIM
0	city	restored_restoration	25.362521	0.777927
1	city	restored_superres	24.973505	0.760798
2	faces	restored_restoration	26.208288	0.817937
3	faces	restored_superres	27.323311	0.810000
4	nature	restored_restoration	24.196838	0.761445
5	nature	restored_superres	25.162843	0.740272

Figure 8: Real-ESRGAN and GPFGAN Models Results on Different Degradations